Homework 1

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ADVANCED MATHEMATICAL PHYSICS

PHYS 721

1.8.1

Goal: Find $\frac{1}{z}$.

proof
$$\frac{1}{z} = \frac{1}{re^{i\theta}} = \frac{1}{r} e^{-i\theta} \quad \text{polar form}$$

$$\frac{1}{r^2} re^{-i\theta} = \frac{1}{x^2+y^2}\left[r\cos(-\theta) + i r\sin(-\theta)\right]$$
$$= \frac{1}{x^2+y^2}\left[x - iy\right] \quad \text{cartesian form}$$

1.8.2

Goal: $z^{1/2} \in \mathbb{C}$ & $i^{1/2} = ?$

proof Let $z = re^{i(\theta+2\pi n)}$

$$z^{1/2} = \left[re^{i(\theta+2\pi n)}\right]^{1/2} = r^{1/2} e^{i(\theta/2 + n\pi)}$$

This leads to 2 unique square roots: $z_1 = r^{1/2} e^{i(\theta/2)}$ & $z_2 = r^{1/2} e^{i(\theta/2 + \pi)}$

Let $z = i = 1e^{i(\pi/2 + 2\pi n)}$

$\Rightarrow$ Then, $z^{1/2} = i^{1/2}$ has 2 roots

$$z_1 = e^{i(\pi/4)} \quad \text{and} \quad z_2 = e^{i(\pi/4 + \pi)} = e^{i(5\pi/4)}$$
$$= \frac{1}{\sqrt{2}}(1+i) \qquad\qquad = -\frac{1}{\sqrt{2}}(1+i)$$

1.8.3

Goal: Find $\cos n\theta$ & $\sin n\theta$ in terms of $\cos m\theta$ & $\sin m\theta$'s

proof
$$\cos n\theta = Re\{(\cos\theta + i\sin\theta)^n\}$$
$$\sin n\theta = Im\{(\cos\theta + i\sin\theta)^n\}$$

$$(\cos\theta + i\sin\theta)^n = \sum_{k=0}^{n} \binom{n}{k} (\cos\theta)^{n-k} (i\sin\theta)^k$$

Even k gives us the real components:

$$\cos n\theta = \cos^n\theta - \binom{n}{2}\cos^{n-2}\theta\sin^2\theta + \binom{n}{4}\cos^{n-4}\theta\sin^4\theta + \dots$$

Odd k gives us the imaginary components:

$$\sin n\theta = \binom{n}{1}\cos^{n-1}\theta\sin\theta - \binom{n}{3}\cos^{n-3}\theta\sin^3\theta + \dots$$

1.8.4 Prove that

(a) $\displaystyle\sum_{n=0}^{N-1} \cos nx = \frac{\sin(Nx/2)}{\sin x/2}\cos(N-1)\frac{x}{2},$

(b) $\displaystyle\sum_{n=0}^{N-1} \sin nx = \frac{\sin(Nx/2)}{\sin x/2}\sin(N-1)\frac{x}{2}.$

1.8.4

a) $$\sum_{n=0}^{N-1} \cos nx = \sum_{n=0}^{N-1} Re\{e^{inx}\} = Re\left\{\sum_{n=0}^{N-1}(e^{ix})^n\right\} = Re\left\{\frac{1-e^{iNx}}{1-e^{ix}}\right\} = Re\left\{\frac{e^{iNx/2}\left(e^{-iNx/2}-e^{iNx/2}\right)}{e^{ix/2}\left(e^{-ix/2}-e^{ix/2}\right)}\right\}$$

$$= Re\left\{e^{i\frac{N-1}{2}x}\cdot\frac{\sin(Nx/2)}{\sin(x/2)}\right\} = \cos\left(\frac{N-1}{2}x\right)\frac{\sin(Nx/2)}{\sin(x/2)}$$

a) $$\sum_{n=0}^{N-1} \sin nx = \sum_{n=0}^{N-1} Im\{e^{inx}\} = Im\left\{\sum_{n=0}^{N-1}(e^{ix})^n\right\} = Im\left\{\frac{1-e^{iNx}}{1-e^{ix}}\right\} = Im\left\{\frac{e^{iNx/2}\left(e^{-iNx/2}-e^{iNx/2}\right)}{e^{ix/2}\left(e^{-ix/2}-e^{ix/2}\right)}\right\}$$

$$= Im\left\{e^{i\frac{N-1}{2}x}\cdot\frac{\sin(Nx/2)}{\sin(x/2)}\right\} = \sin\left(\frac{N-1}{2}x\right)\frac{\sin(Nx/2)}{\sin(x/2)}$$

1.8.5

$$\sinh(iz) = \frac{e^{iz}-e^{-iz}}{2} = i\,\frac{e^{iz}-e^{-iz}}{2i} = i\sin z$$

$$\sin(iz) = \frac{e^{-z}-e^{z}}{2i} = i\,\frac{e^{z}-e^{-z}}{2} = i\sinh z$$

$$\cosh(iz) = \frac{e^{iz}+e^{-iz}}{2} = \cos z$$

$$\cos(iz) = \frac{e^{-z}+e^{z}}{2} = \cosh z$$

1.8.6 Using the identities

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \qquad \sin z = \frac{e^{iz} - e^{-iz}}{2i},$$

established from comparison of power series, show that

(a) $\sin(x + iy) = \sin x \cosh y + i \cos x \sinh y,$

$\cos(x + iy) = \cos x \cosh y - i \sin x \sinh y,$

(b) $|\sin z|^2 = \sin^2 x + \sinh^2 y, \quad |\cos z|^2 = \cos^2 x + \sinh^2 y.$

This demonstrates that we may have $|\sin z|, |\cos z| > 1$ in the complex plane.

1.8.6

a) $\sin x \cosh y + i \cos x \sinh y$

$$= \frac{e^{ix} - e^{-ix}}{2i} \cdot \frac{e^{y} + e^{-y}}{2} + i\, \frac{e^{ix} + e^{-ix}}{2} \cdot \frac{e^{y} - e^{-y}}{2}$$

$$= \frac{e^{ix}e^{y} - e^{-ix}e^{y} + e^{ix}e^{-y} - e^{-ix}e^{-y}}{4i} + \frac{i}{4}\left[e^{ix}e^{y} + e^{-ix}e^{y} - e^{ix}e^{-y} - e^{-ix}e^{-y}\right]$$

$$= \frac{i}{4}\left[-\cancel{e^{ix}e^{y}} + e^{-ix}e^{y} - e^{ix}e^{-y} + \cancel{e^{-ix}e^{-y}}\right] + \frac{i}{4}\left[\cancel{e^{ix}e^{y}} + e^{-ix}e^{y} - e^{ix}e^{-y} - \cancel{e^{-ix}e^{-y}}\right]$$

$$= \frac{e^{ix}e^{-y} - e^{-ix}e^{y}}{2i} = \frac{e^{i(x+iy)} - e^{-i(x+iy)}}{2i} = \sin(x+iy)$$

$$\cos(x+iy) = \cos x \cos iy - \sin x \sin iy$$

$$= \cos x \cosh y - i \sin x \sinh y \qquad (1.8.5)$$

b) $|\sin z|^2 = \sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y$

$$= \sin^2 x \left[1 + \cancel{\sinh^2 y}\right] + \left[1 - \cancel{\sin^2 x}\right] \sinh^2 y = \sin^2 x + \sinh^2 y$$

$|\cos z|^2 = \cos^2 x \cosh^2 y + \sin^2 x \sinh^2 y$

$$= \cos^2 x \left[1 + \cancel{\sinh^2 y}\right] + \left[1 - \cancel{\cos^2 x}\right] \sinh^2 y = \cos^2 x + \sinh^2 y$$

1.8.7 From the identities in Exercises 1.8.5 and 1.8.6 show that

(a) $\sinh(x+iy) = \sinh x \cos y + i \cosh x \sin y$,

$\cosh(x+iy) = \cosh x \cos y + i \sinh x \sin y$,

(b) $|\sinh z|^2 = \sinh^2 x + \sin^2 y$, $|\cosh z|^2 = \cosh^2 x + \sin^2 y$.

a) $\frac{i}{i}\sinh(x+iy) = \frac{\sin(ix-y)}{i} = \frac{1}{i}\left[\underset{\text{odd}}{\sin(-y)}\cosh(x) + i\underset{\text{even}}{\cos(-y)}\sinh(x)\right]$

$= \sinh x \cos y + i \cosh x \sin y$

$\cosh(x+iy) = \cos(-y+ix) = \cos(-y)\cosh(x) - i\sin(-y)\sinh(x)$

$= \cosh x \cos y + i \sinh x \sin y$

b) $|\sinh z|^2 = \sinh^2 x \cos^2 y + \cosh^2 x \sin^2 y$

$= \sinh^2 x\,[1-\cancel{\sin^2 y}] + [1+\cancel{\sinh^2 y}]\sin^2 y = \sinh^2 x + \sin^2 y$

$|\cosh z|^2 = \cosh^2 x \cos^2 y + \sinh^2 x \sin^2 y$

$= \cosh^2 x\,[1-\cancel{\sin^2 y}] + [\cancel{\cosh^2 x} - 1]\sin^2 y = \cosh^2 x - \sin^2 y$

1.8.8 Show that

(a) $\tanh\frac{z}{2} = \frac{\sinh x + i \sin y}{\cosh x + \cos y}$, (b) $\coth\frac{z}{2} = \frac{\sinh x - i \sin y}{\cosh x - \cos y}$.

a) $\tanh\frac{z}{2} = \frac{e^{z/2} - e^{-z/2}}{e^{z/2} + e^{-z/2}} = \frac{e^z - 1}{e^z + 1} = \frac{e^x e^{iy} - 1}{e^x e^{iy} + 1}$ $/e^x$ $/e^x$

$= \frac{\cos y + i \sin y - e^{-x}}{\cos y + i \sin y + e^{-x}}$ · multiply by conjugate of denominator

$= \frac{(\cos y - e^{-x})(\cos y + e^{-x}) + \sin^2 y + i \sin y\,(\cos y + e^{-x} - \cos y + e^{-x})}{(e^{-x} + \cos y)^2 + \sin^2 y}$

$$= \frac{\cos^2 x - e^{-2x} + \sin^2 x + 2ie^{-x}\sin y}{e^{-2x} + 2e^{-x}\cos y + \cos^2 y + \sin^2 y}$$

$\cdot\, e^x$

$\cdot\, e^x$

$$= \frac{e^x - e^{-x} + 2i\sin y}{e^x + e^{-x} + 2\cos y} = \frac{\sinh x + i\sin y}{\cosh x + \cos y}$$

b) $$\coth\frac{z}{2} = \frac{e^{z/2} + e^{-z/2}}{e^{z/2} - e^{-z/2}} = \frac{e^z + 1}{e^z - 1} = \frac{e^x e^{iy} + 1}{e^x e^{iy} - 1} \quad \begin{matrix} /e^x \\ /e^x \end{matrix}$$

$$= \frac{\cos y + i\sin y + e^{-x}}{\cos y + i\sin y - e^{-x}}$$

$\cdot$ multiply by conjugate of denominator

$$= \frac{(\cos y - e^{-x})(\cos y + e^{-x}) + \sin^2 y + i\sin y(-\cos y - e^{-x} + \cos y - e^{-x})}{(-e^{-x} + \cos y)^2 + \sin^2 y}$$

$$= \frac{e^x - e^{-x} - 2i\sin y}{e^x + e^{-x} - 2\cos y} = \frac{\sinh x - i\sin y}{\cosh x - \cos y}$$

1.8.9 By comparing series expansions, show that $\tan^{-1} x = \frac{i}{2}\ln\left(\frac{1-ix}{1+ix}\right)$.

$$\tan^{-1} x = \int \frac{1}{1+x^2}\,dx$$

$$= \int \sum_{n=0}^{\infty} (-x^2)^n\,dx$$

$$= \int \sum_{n=0}^{\infty} ((ix)^2)^n\,dx$$

$$= \int 1 + i^2x^2 + i^4x^4 + \dots\,dx$$

$$= \tfrac{1}{2}\int 1 + ix + i^2x^2 + i^3x^3 + i^4x^4 + \dots\,dx$$

$$+ \tfrac{1}{2}\int 1 - ix + i^2x^2 - i^3x^3 + i^4x^4 + \dots\,dx$$

$$= \tfrac{1}{2}\left[\int \frac{1}{1-ix}\,dx + \int \frac{1}{1+ix}\,dx\right]$$

$$= \tfrac{1}{2}\left[-\tfrac{1}{i}\ln(1-ix) + \tfrac{1}{i}\ln(1+ix)\right]$$

$$= \tfrac{i}{2}\ln\left(\frac{1+ix}{1-ix}\right)$$

1.8.10 Find the Cartesian form for **all values** of

(a) $(-8)^{1/3}$,

(b) $i^{1/4}$,

(c) $e^{i\pi/4}$.

1.8.11 Find the polar form for **all values** of

(a) $(1+i)^3$,

(b) $(-1)^{1/5}$.

1.8.10

a) $(-8)^{1/3} = \left[8e^{i(\pi+2\pi n)}\right]^{1/3}$ for $n = 0, 1, 2$

$\Rightarrow \quad 2e^{i(\pi/3)} = 2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = 1 + i\sqrt{3}$

$2e^{i\pi} = -2$

$2e^{i(5\pi/3)} = 2\left(\frac{1}{2} - i\frac{\sqrt{3}}{2}\right) = 1 - i\sqrt{3}$

b) $i^{1/4} = \left[1e^{i(\pi/2 + 2\pi n)}\right]^{1/4}$ for $n = 0, 1, 2, 3$

$\Rightarrow e^{i(\pi/8)}, \; e^{i(5\pi/8)}, \; e^{i(9\pi/8)}, \; e^{i(13\pi/8)}$

$= \frac{1}{2}\left(\sqrt{2+\sqrt{2}} + i\sqrt{2-\sqrt{2}}\right), \; \frac{1}{2}\left(-\sqrt{2-\sqrt{2}} + i\sqrt{2+\sqrt{2}}\right), \; \frac{1}{2}\left(-\sqrt{2+\sqrt{2}} - i\sqrt{2-\sqrt{2}}\right), \; \frac{1}{2}\left(\sqrt{2-\sqrt{2}} - i\sqrt{2+\sqrt{2}}\right)$

* $\cos\pi/8 = \sqrt{\frac{1+\cos\pi/4}{2}} = \frac{1}{2}\sqrt{2+\sqrt{2}}$, $\sin\pi/8 = \sqrt{\frac{1-\cos x}{2}} = \frac{1}{2}\sqrt{2-\sqrt{2}}$

half-angle formulae

c) $e^{i\pi/4} = \cos\pi/4 + i\sin\pi/4$

$= \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$

1.8.11

a) $(1+i)^3 \Rightarrow r = \sqrt{1^2+1^2} = \sqrt{2} \Rightarrow (1+i)^3 = \left(\sqrt{2}e^{i\pi/4}\right)^3 = \sqrt{2}e^{i\,3\pi/4}$

$\theta = \pi/4$

b) $(-1)^{1/5} \Rightarrow r = 1 \Rightarrow (-1)^{1/5} = \left[e^{i(\pi+2\pi n)}\right]^{1/5} = e^{i\left(\pi/5 + \frac{2\pi n}{5}\right)}$ for $n = 0, 1, 2, 3, 4$. unique roots

$\theta = \pi$

Proof 1.137

$$\sin^{-1}(z) = -i\ln\left[iz+\sqrt{1-z^2}\right], \quad \tan^{-1}(z) = \frac{i}{2}\left[\ln(1-iz)-\ln(1+iz)\right],$$
$$\sinh^{-1}(z) = \ln\left[z+\sqrt{1+z^2}\right], \quad \tanh^{-1}(z) = \frac{1}{2}\left[\ln(1+z)-\ln(1-z)\right]. \qquad (1.137)$$

Proving $\sin^{-1} z$:

$$e^{i\theta} = i\sin\theta + \sqrt{1-\sin^2\theta}$$

Let $z = \sin\theta$ & taking ln on both sides:

$$\theta = \sin^{-1} z = -i\ln\left[iz+\sqrt{1-z^2}\right]$$

Proving $\sinh^{-1} z$:

$$e^{\theta} = \sinh\theta + \cosh\theta$$
$$e^{\theta} = \sinh\theta + \sqrt{1+\sinh^2\theta}$$

Let $z = \sinh\theta$ & taking ln on both sides :

$$\theta = \ln\left[z+\sqrt{1+z^2}\right]$$

Proving $\tan^{-1} z$:

$$e^{2i\theta} = \frac{1+i\tan\theta}{1-i\tan\theta} \qquad (1)$$

Let $z = \tan\theta$ & taking ln on both sides :

$$\theta = \frac{i}{2}\left[\ln(1-iz)-\ln(1+iz)\right]$$

Proving $\tanh^{-1} z$:

$$e^{2\theta} = \frac{1+\tanh\theta}{1-\tanh\theta} \qquad (2)$$

Let $z = \tanh\theta$ & taking ln on both sides :

$$\theta = \frac{1}{2}\left[\ln(1+z)-\ln(1-z)\right]$$

$$\tan\theta = \frac{e^{i\theta}-e^{-i\theta}}{i(e^{i\theta}+e^{-i\theta})}$$
$$\frac{e^{2i\theta}-1}{i(e^{2i\theta}+1)}$$
$$i\tan\theta\,(e^{2i\theta}+1) = e^{2i\theta}-1$$
$$(i\tan\theta-1)\,e^{2i\theta} = -(1+i\tan\theta)$$
$$e^{2i\theta} = \frac{1+i\tan\theta}{1-i\tan\theta} \qquad (1)$$

$$\tanh\theta = \frac{e^{\theta}-e^{-\theta}}{e^{\theta}+e^{-\theta}}.$$
$$= \frac{e^{2\theta}-1}{e^{2\theta}+1}$$
$$\tanh\theta\,(e^{2\theta}+1) = e^{2\theta}-1$$
$$(\tanh\theta-1)\,e^{2\theta} = -(1+\tanh\theta)$$
$$e^{2\theta} = \frac{1+\tanh\theta}{1-\tanh\theta} \qquad (2)$$

11.2.1 Show whether or not the function $f(z) = \Re(z) = x$ is analytic.

11.2.1

Let f be analytic, then f satisfies the cauchy - Reimann conditions.

$f(z) = Re\{z\} = x \Rightarrow u(x,y) = x, \; v(x,y) = 0$

CRC

$\partial u_x = 1 \neq 0 = \partial v_y \Rightarrow$ f(z) is not analytic anywhere in its domain

11.2.2 Having shown that the real part $u(x, y)$ and the imaginary part $v(x, y)$ of an analytic function $w(z)$ each satisfy Laplace's equation, show that neither $u(x, y)$ nor $v(x, y)$ **can have either a maximum or a minimum** in the interior of any region in which $w(z)$ is analytic. (They can have saddle points only.)

proof

WLOG. We work with the function $u(x,y)$.

Suppose $u(x,y)$ satisfies the Laplace equation i.e.

$U_{xx} + U_{yy} = 0$

Suppose $u(x,y)$ has a maximum or minimum at (a,b). Then,

we require 2 conditions:

1) First derivatives must be 0: $U_x(a,b) = 0, \; U_y(a,b) = 0$

2) Discriminant must be positive: $D = U_{xx} U_{yy} - (U_{xy})^2 > 0$

We investigate the 2nd condition:

$$D = U_{xx} U_{yy} - (U_{xy})^2$$

$$= -(U_{yy})^2 - (U_{xy})^2 < 0 \qquad U_{xx} = -U_{yy} \text{ from Laplace's Equation}$$

D is therefore negative as long as U_{yy} & $U_{xy} \neq 0$. However, if they are equal to zero. $U_{xx} = U_{yy} = U_{xy} = U_x = U_y = 0$. Then, $u(x,y) = c$ for some constant and therefore will not have a max nor min. Thus, $u(x,y)$ must not have a max or min.

The same argument can be done for $v(x,y)$

11.2.3 Find the analytic function

$$w(z) = u(x, y) + iv(x, y)$$

(a) if $u(x, y) = x^3 - 3xy^2$, (b) if $v(x, y) = e^{-y} \sin x$.

a) If w is analytic $\Rightarrow \partial_x u = \partial_y v$ (1) & $\partial_y u = -\partial_x v$ (2)

(1) $v = \int \partial_x u \, dy = \int 3x^2 - 3y^2 \, dy = 3x^2 y - y^3 + g(x)$

(2) $v = -\int \partial_y u \, dx = -\int -6xy \, dx = 3x^2 y + h(y)$

$g(x) = 0$

$h(y) = -y^3$.

$\Rightarrow v(x, y) = 3x^2 y - y^3$ for $w(z)$ to be analytic.

b) If w is analytic $\Rightarrow \partial_x u = \partial_y v$ (1) & $\partial_y u = -\partial_x v$ (2)

(1) $u = \int \partial_y v \, dx = \int -e^{-y} \sin x \, dx = e^{-y} \cos x + g(y)$

(2) $u = \int -\partial_x v \, dy = \int -e^{-y} \cos x \, dy = e^{-y} \cos x + h(x)$

$g(y) = h(x) = 0$.

$\Rightarrow u(x, y) = e^{-y} \cos x$ for $w(z)$ to be analytic.

11.2.4 If there is some common region in which $w_1 = u(x, y) + iv(x, y)$ and $w_2 = w_1^* = u(x, y) - iv(x, y)$ are both analytic, prove that $u(x, y)$ and $v(x, y)$ are constants.

<u>proof</u> CRC for w_1 : $U_x = V_y$, $U_y = -V_x$

CRC for w_2 : $U_x = -V_y$, $U_y = V_x$

Suppose $\exists$ a region such that CRC satisfy both w_1 & w_2, then

$U_x = U_y = V_x = V_y = 0$ $\Rightarrow$ All first order derivatives of u & v are zero $\Rightarrow$

u & v must be constant on the whole domain.

11.2.5 Starting from $f(z) = 1/(x+iy)$, show that $1/z$ is analytic in the entire finite z plane except at the point $z = 0$. This extends our discussion of the analyticity of z^n to negative integer powers n.

Let $f(z) = \frac{1}{x+iy} = \frac{x-iy}{x^2+y^2} \Rightarrow u(x,y) = \frac{x}{x^2+y^2}$ and $v(x,y) = \frac{-y}{x^2+y^2}$

Let us check if $f(z)$ satisfies CRC :

(1) $u_x = \frac{x^2+y^2-2x^2}{(x^2+y^2)^2} = \frac{-(x^2+y^2)+2y^2}{(x^2+y^2)^2} = v_y$, as long as $x^2+y^2 \neq 0$ i.e. $z \neq 0$

(2) $u_y = \frac{-2xy}{(x^2+y^2)^2} = -\frac{2xy}{(x^2+y^2)^2} = -v_x$, as long as $x^2+y^2 \neq 0$ i.e. $z \neq 0$

$f(z)$ satisfies CRC for all values of $z \neq 0 \Rightarrow f(z)$ is analytic everywhere except $z = 0$.

11.2.6 Show that given the Cauchy-Riemann equations, the derivative $f'(z)$ has the same value for $dz = a\,dx + ib\,dy$ (with neither a nor b zero) as it has for $dz = dx$.

Goal :

Show that f' is equivalent for $dz = a\,dx + i\,b\,dy$ & $dz = dx$

proof

Let f be a complex valued function s.t. $f(x,y) = u(x,y) + i\,v(x,y)$

The total differential $df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$

$$= (u_x + iv_x)\,dx + (u_y + iv_y)\,dy$$

Now, if we are to take the directional derivative $dz = a\,dx + i\,b\,dy$ s.t. $\Delta x = a\,dx$ & $\Delta y = b\,dy$. Then,

$$\frac{df}{dz} = \frac{(u_x+iv_x)\,a\,dx + (u_y+iv_y)\,b\,dy}{a\,dx + ib\,dy}$$

$$= \frac{(u_x+iv_x)\,a\,dx + (-v_x+iu_x)\,b\,dy}{a\,dx + ib\,dy}$$

$$= \frac{(u_x+iv_x)\,\cancel{a\,dx + (u_x+iv_x)\,ib\,dy}}{\cancel{a\,dx + ib\,dy}} = u_x + iv_x = \frac{df}{dx} \text{ as required.}$$

11.2.7 Using $f(re^{i\theta}) = R(r,\theta)e^{i\Theta(r,\theta)}$, in which $R(r,\theta)$ and $\Theta(r,\theta)$ are differentiable real functions of r and θ, show that the Cauchy-Riemann conditions in polar coordinates become

$$\text{(a)} \quad \frac{\partial R}{\partial r} = \frac{R}{r}\frac{\partial \Theta}{\partial \theta}, \qquad \text{(b)} \quad \frac{1}{r}\frac{\partial R}{\partial \theta} = -R\frac{\partial \Theta}{\partial r}.$$

Hint. Set up the derivative first with δz radial and then with δz tangential.

Analogous to the proof given for the cartesian coordinates $\Rightarrow$

Let $z = re^{i\theta}$ & $f(z) = R(r,\theta)\, e^{i\Theta(r,\theta)}$

Then,

$$\delta z = e^{i\theta}\,\delta r + ire^{i\theta}\,\delta\theta$$

$$\delta f = e^{i\Theta}\,\delta R + iRe^{i\Theta}\,\delta\Theta$$

Therefore, $$\frac{\delta f}{\delta z} = \frac{e^{i\Theta}\,\delta R + iRe^{i\Theta}\,\delta\Theta}{e^{i\theta}\,\delta r + ire^{i\theta}\,\delta\theta}$$

Taking $\frac{\delta f}{\delta z}$ in the radial direction & $\displaystyle\lim_{\delta r \to 0} \frac{\delta f}{\delta z} = \frac{e^{i\Theta}}{e^{i\theta}}\left[-\frac{i}{r}R_\theta + \frac{1}{r}R\,\Theta_\theta\right]$

Taking $\frac{\delta f}{\delta z}$ in the tangential direction & $\displaystyle\lim_{\delta\theta \to 0} \frac{\delta f}{\delta z} = \frac{e^{i\Theta}}{e^{i\theta}}\left[R_r + iR\,\Theta_r\right]$

Since the derivative must be consistent in both directions &

(a) $R_r = \frac{1}{r}R\,\Theta_\theta$ (b) $\frac{1}{r}R_\theta = -R\,\Theta_r$ as required.

11.2.8 As an extension of Exercise 11.2.7 show that $\Theta(r,\theta)$ satisfies the 2-D Laplace equation in polar coordinates,

$$\frac{\partial^2\Theta}{\partial r^2} + \frac{1}{r}\frac{\partial\Theta}{\partial r} + \frac{1}{r^2}\frac{\partial^2\Theta}{\partial\theta^2} = 0.$$

We attempt to find the 2-D Laplace Equation for Θ by trying to find the 2^{nd} order derivatives of Θ through the polar CRC

(a) $R_r = \frac{1}{r} R\Theta_\theta$ (b) $\frac{1}{r} R_\theta = -R\Theta_r$

Taking $\frac{1}{\partial\theta}$ of (a) : $R_{r\theta} = \frac{1}{r}[R_\theta \Theta_\theta + R\Theta_{\theta\theta}]$

Taking $\frac{1}{\partial r}$ of (b) : $-\frac{1}{r^2} R_\theta + \frac{1}{r} R_{r\theta} = -R_r\Theta_r - R\Theta_{rr}$

Substituting $R_{r\theta}$ from (1) to (2) :

$-\frac{1}{r^2} R_\theta + \frac{1}{r}\frac{1}{r}[R_\theta\Theta_\theta + R\Theta_{\theta\theta}] = -R_r\Theta_r - R\Theta_{rr}$

Substituting R_r and R_θ from (a) & (b) :

$\frac{1}{r}R\Theta_r + \frac{1}{r^2}[\cancel{-rR\Theta_r\Theta_\theta} + R\Theta_{\theta\theta}] = \cancel{-\frac{1}{r}R\Theta_\theta\Theta_r} - R\Theta_{rr}$

$\frac{1}{r}R\Theta_r + \frac{1}{r^2}R\Theta_{\theta\theta} = -R\Theta_{rr}$

$\Theta_{rr} + \frac{1}{r}\Theta_r + \frac{1}{r^2}\Theta_{\theta\theta} = 0$ ▨

11.2.9 For each of the following functions $f(z)$, find $f'(z)$ and identify the maximal region within which $f(z)$ is analytic.

(a) $f(z) = \dfrac{\sin z}{z}$, (d) $f(z) = e^{-1/z}$,

(b) $f(z) = \dfrac{1}{z^2+1}$, (e) $f(z) = z^2 - 3z + 2$,

(c) $f(z) = \dfrac{1}{z(z+1)}$, (f) $f(z) = \tan(z)$,

(g) $f(z) = \tanh(z)$.

a) $f'(z) = \dfrac{\cos z}{z} - \dfrac{\sin z}{z^2}$

potential problems :

✓ $z \to 0$: $\lim\limits_{z\to 0} f'(z) = \lim\limits_{z\to 0} \sin z - \dfrac{\sin z}{z} = 0$ (L'hopital)

✗ $z \to \infty$: $\lim\limits_{z\to\infty} f'(z) = \lim\limits_{z\to\infty} \sin z - \dfrac{\sin z}{z} = \text{DNE}$

Analytic everywhere except at $z \to \infty$.

b) $f'(z) = \dfrac{-2z}{(z^2+1)^2}$

potential problems :

✗ $z \to \pm i$: $\lim\limits_{z\to\pm i} f'$ DNE

✓ $z \to \infty$: $\lim\limits_{z\to\infty} f' = 0$

Analytic everywhere except at $z = \pm i$

c) $f'(z) = -\dfrac{(2z+1)}{z^2(z+1)^2}$

Analytic everywhere except $z = 0, -1$

d) $f'(z) = \frac{1}{z^2} e^{-1/z}$

Analytic everywhere except at $z = 0$.

e) $f'(z) = 2z - 3$

Analytic everywhere except at $z \to \infty$

f) $f'(z) = \sec^2 z$

$= \dfrac{1}{\cos^2 z}$

Analytic everywhere except at $z \to \infty$ and roots z_n of $\cos z_n = 0$.

g) $f'(z) = \operatorname{sech}^2 z$

$= \dfrac{1}{\cosh^2 z}$

Analytic everywhere except at $z \to \infty$ and roots z_n of $\cosh z_n = 0$.

11.2.10 For what complex values do each of the following functions $f(z)$ have a derivative?

(a) $f(z) = z^{3/2}$,
(b) $f(z) = z^{-3/2}$,
(c) $f(z) = \tan^{-1}(z)$,
(d) $f(z) = \tanh^{-1}(z)$.

For parts a & b, while we can use the power rule to find derivatives
These function are multivalued! i.e. it would not make sense to talk about a derivative
unless we restrict the domain. We will the details for later but for
now.

a) & b) f' exist $\forall z$ except $z = 0$.

c) $\frac{d}{dz}(\tan^{-1} z) = \frac{1}{1+z^2}$ $\Rightarrow$ f' exists everywhere except $z = \pm i$

d) $\frac{d}{dz}(\tanh^{-1} z) = \frac{1}{1-z^2}$ $\Rightarrow$ f' exists everywhere except $z = \pm 1$

↑ proof Let $w = \tanh z$
$w' = \operatorname{sech}^2 z$
$w' = 1 - \tanh^2 z$

$\operatorname{sech}^2 z = \frac{1}{\cosh^2 z} = \frac{\cosh^2 z - \sinh^2 z}{\cosh^2 z}$
$= 1 - \tanh^2 z$

$\frac{d}{dx} f^{-1}(x) = \frac{1}{f'(f^{-1}(x))}$

$\frac{d}{dz}(\tanh^{-1} z) = \frac{1}{1 - \tanh^2(\tanh^{-1} z)} = \frac{1}{1-z^2}$.

11.2.11 Two-dimensional irrotational fluid flow is conveniently described by a complex potential $f(z) = u(x, v) + iv(x, y)$. We label the real part, $u(x, y)$, the velocity potential, and the imaginary part, $v(x, y)$, the stream function. The fluid velocity $\mathbf{V}$ is given by $\mathbf{V} = \nabla u$. If $f(z)$ is analytic:

(a) Show that $df/dz = V_x - iV_y$.
(b) Show that $\nabla \cdot \mathbf{V} = 0$ (no sources or sinks).
(c) Show that $\nabla \times \mathbf{V} = 0$ (irrotational, nonturbulent flow).

a) from 11.2.6 : $\frac{df}{dz} = \frac{df}{dx} = u_x + i v_x = u_x - i u_y = V_x - i V_y$

b) Goal : $\vec{\nabla} \cdot \vec{V} = 0$

proof

$\vec{\nabla} \cdot \vec{\nabla} u = \vec{\nabla} \cdot (u_x, u_y) = u_{xx} + u_{yy} = 0$ Laplace Equation (11.13)

c) Goal : $\vec{\nabla} \times \vec{V} = 0$

proof

$\vec{\nabla} \times \vec{\nabla} u = 0$ The curl of a scalar field is 0.

$\vec{\nabla} \times (u_x, u_y)$

$= u_{yx} - u_{xy} = 0$ The derivative operator is commutative here.

11.2.12 The function $f(z)$ is analytic. Show that the derivative of $f(z)$ with respect to z^* does not exist unless $f(z)$ is a constant.

Hint. Use the chain rule and take $x = (z + z^*)/2$, $y = (z - z^*)/2i$.

Note. This result emphasizes that our analytic function $f(z)$ is not just a complex function of two real variables x and y. It is a function of the complex variable $x + iy$.

Goal: Show that $\frac{df}{dz^*}$ satisfy CRC only when $f(z) = c$ constant

Proof

Let's try to find CRC for the derivative of $f(z)$ in the z^* direction

$$\lim_{\delta y \to 0} \frac{\delta f}{\delta z^*} = u_x + i v_x = i u_y - v_y = \lim_{\delta x \to 0} \frac{\delta f}{\delta z^*}$$

derivative in x-direction

derivative in y-direction

We get that

$$\left.\begin{aligned} u_x &= -v_y \\ v_y &= v_x \end{aligned}\right\}$$

These are CRC but in the opposite signs!

These cannot hold unless $u_x = u_y = v_x = v_y = 0$

$\Rightarrow f$ is constant.